

AURICULARIALES MAP

Steps to making the first map, showing mushroom distribution (of the order Auriculariales) by hydrobasins in Costa Rica

1. Create a new map called Auriculariales, save to your drive, and make a new folder for this project
2. From the Project subribbon, add the Emilyleva_Datasets folder
3. From the Catalog pane, add the 'country.shp' shapefile to your map
 - a. This file should be located through this pathway Folders → mgisdata → World → country.shp
4. In the contents pane, right click the 'country' feature layer and open the attribute table
5. Highlight (what should be the row for Costa Rica) row 43
 - a. With the row still highlighted, click the 'Table' ribbon, then 'Export Features' in the Export subribbon
 - b. In Parameters, make sure the Input Feature is 'country' and that 'Use the selected records: 1' is toggled on
 - c. For the Output Feature Class, click the folder icon to browse your files, locate the project database and save the feature class there as "cr_boundary"
 - d. Click 'OK'
 - e. Turn off and remove 'country' layer once 'cr_boundary' has successfully loaded into the Contents pane
6. In the Contents pane, right click the cr_boundary layer and choose Zoom To Layer
7. To correctly symbolize the country boundary, click on the color box underneath the 'cr_boundary' layer to open the Symbology pane
 - a. Choose the Black Outline (2pt) option in the Gallery
 - b. Then switch to the Properties tab and under Appearance, change the Outline Width to 3 pt

8. Go back to the Catalog Pane and add the mushroom distribution table, 'CR_auriculariales.csv'
 - a. This file should be located through this pathway Folders → Emily → CR_auriculariales.csv → CR_auriculariales.csv
9. In the Contents Pane under 'Standalone Tables', right click CR_auriculariales.csv and from 'Create Points From Table' click 'XY Table to Point'
 - a. Input table should be CR_auriculariales.csv
 - b. For the Output Feature Class, click the folder icon to browse your files, locate the project database and save the feature class there as 'auricularia_dist'
 - c. The X Field should be longitude and the Y Field should be latitude
 - d. Don't change the CS
10. In the Geoprocessing Pane, search for the 'Delete Field' tool from the Data Management toolbox
 - a. Set the Input Table as auricularia_dist
 - b. Then under Methods, use the down arrow to choose 'Keep Fields'
 - c. Under fields check the boxes for id, url, latitude, longitude, common_name, and scientific_name
 - d. Click 'Run'

Next, add in the hydrobasins

11. Go to the Catalog Pane and add the 'hydrobasins_centralam.shp' shapefile to the map
 - a. This file should be located through this pathway Folders → Emily → Datasets → hydrobasins_centralam → hydrobasins_centralam.shp
12. Since this shapefile represents hydrobasins in all of Central America, we need to clip it to just Costa Rica. Go back to the Geoprocessing Pane and find the 'Clip' tool
 - a. For Input Features or Dataset, use the drop down arrow to choose 'hydrobasins_centralam' and for Clip Feature choose the 'cr_boundary' layer

- b. Save the new clipped layer to the project geodatabase, but be sure to name it 'hydrobasins'
 - c. Click 'Run'
 - d. Once the hydrobasins layer has successfully loaded into the Contents Pane, you can remove the 'hydrobasins_centralam' layer
13. At this point, make sure that the cr_boundary layer is below hydrobasins, which should be below the auricularia_dist layer

Symbolize the hydrobasins

14. Click on the color box underneath the hydrobasins layer to open the Symbolology for the layer
- a. Click the circled back arrow above the 'Gallery' tab to go to the Primary symbology section.
 - b. Right underneath the section name use the drop down arrow to choose Graduated Color symbology instead of Single Symbol
 - c. Change the Field to SUB_AREA with 5 classes
 - d. Choose the Cyan-Light to Blue-Dark color scheme
15. To symbolize the mushroom point data, click on the dot symbol underneath the auricularia_dist layer to open the layer's Symbolology pane
- a. From the Gallery, choose Tear Pin 1 (40%)
 - b. Switch to the Properties tab
 - c. Under the Appearance ribbon, choose a nice yellow color
 - d. For the outline color, choose a medium shade of gray and make the Outline Width 1pt
 - e. Click 'Apply'

Add a new field

16. Right click the auricularia_dist layer and open the attribute table
17. Next to 'Field' at the top of the attribute table, click the 'Add' button to add the new field
 - a. Name the new field "fungal_order" and NOT "order" as it is apparently an illegal word in ArcGIS Pro
 - b. Then make sure that the Data Type is text
 - c. Go to the top of the screen and choose 'Save' within the Manage Edits ribbon
 - d. Exit out of the Fields tab
18. In the regular attribute table tab, find the new field you created
 - a. Right click the top/the field title where it says 'fungal_order' and choose 'Calculate Field'
 - b. In the Calculate Field tool go to the text box underneath "fungal_order =" and write the following exactly
 - i. 'Auriculariales' (include the apostrophes!!)
 - c. Click 'Apply' to make sure it puts Auriculariales into each box underneath the new field, and then 'OK'

Use the Summarize Within Tool

19. Open the Geoprocessing pane and search for the Summarize Within tool from the Analysis toolbox
 - a. For the Input Polygons, use the dropdown arrow to select the hydrobasins layer
 - b. Select the auricularia_dist layer for the Input Summary Features
 - c. Save the Output Feature Class to the project geodatabase and name it 'fungi_by_basin'
 - d. Use the dropdown arrow to select fungal_order as the group field
 - e. Click 'Run'

20. Once the fungi_by_basin layer has successfully loaded into the Contents pane, turn off the hydrobasins layer

- a. Furthermore, drag the hydrobasins layer below the cr_boundary layer for nicer organization

Project everything mapped so far

21. In the Content Pane, right click the map (top of the pane if listed by drawing order) and go to Properties.

- a. Find the Coordinate Systems tab, then in the search box, type “SIRGAS 2000 UTM Zone 16” then press ‘Enter’ on the keyboard
- b. Make sure that you select the same projection typed, then click on the star next to it in order to save it to Favorites
- c. Click ‘OK’

22. Now search for and open the Project tool in the Geoprocessing Pane

- a. For Input Dataset or Feature Class, use the dropdown arrow to select the cr_boundary layer
- b. Click on the folder icon next to Output Dataset or Feature Class, then name the file ‘prj_boundary’ and save it in the project geodatabase
- c. Use the dropdown arrow to select Current Map for the Output Coordinate System
- d. Keep the transformation, then click ‘Run’
- e. Once the newly projected layer successfully loads into the Contents Pane, remove the original layer ‘cr_boundary’, but leave the Project tool open

23. In the Project tool, change the Input Dataset to fungi_by_basin and name the Output Dataset ‘prj_fbb’

- a. Make sure that the Output CS is to be saved in the project geodatabase, then click ‘Run’
- b. Once the newly projected layer successfully loads into the Contents Pane, remove the original layer ‘fungi_by_basin’ and ‘hydrobasins’

24. In the Project tool, change the Input Dataset to auricularia_dist and name the Output Dataset 'prj_auricularia'

- a. Make sure that the Output CS is to be saved in the project geodatabase, then click 'Run'
- a. Once the newly projected layer successfully loads into the Contents Pane, remove the original layer 'auricularia_dist'

Separate the hydrobasin polygons with the highest counts of auriculariales observed

25. In the Contents pane, right click the prj_fbb layer and open the attribute table

- a. Select rows 3, 17, 20, 26, and 27

26. At the top of the screen, click Export Features from within the Export subribbon

- a. For Input Features, prj_fbb should already be selected
- b. 'Use the selected records: 5' should also be selected
- c. Save the Output Feature Class to the project geodatabase and name it 'dense_basins'
- d. Click 'OK'

Symbolize the new feature layer

27. Right click the new layer in the Contents pane and choose Symbology

- a. Change the symbology to Single Symbol
- b. Click the color box next to Symbol and go to the Gallery tab
- c. Choose the Black Outline (1pt), then go back to the Properties tab
- d. Change the Outline Color to Poinsettia Red and change the outline to 1.5pt
- e. Click 'Apply'

Project the new feature layer

28. Search for and open the Project tool in the Geoprocessing Pane
 - a. For Input Dataset or Feature Class, use the dropdown arrow to select the dense_basins layer
 - b. Click on the folder icon next to Output Dataset or Feature Class, then name the file 'prj_dense_basins' and save it in the project geodatabase
 - c. Use the dropdown arrow to select Current Map for the Output Coordinate System
 - d. Keep the transformation, then click 'Run'
 - e. Once the newly projected layer successfully loads into the Contents Pane, remove the original layer 'dense_basins'
 29. Save!
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HELOTIALES MAP

Repeating all of the same steps as the previous map, but this time for the second fungal order, Helotiales

1. Go to Insert, then New Map to add a new map tab to the project
 - a. Title this map 'Helotiales'
2. In the Catalog pane, go to the project geodatabase under the Databases folder and add the following layers
 - a. cr_boundary
 - b. hydrobasins
2. Still in the the Catalog pane, add the mushroom distribution table, 'CR_helotiales.csv'
 - a. This file should be located through this pathway Folders → Emilyleva_Datasets → CRhelotiales.csv → CR_helotiales.csv
3. To correctly symbolize the country boundary, click on the color box underneath the 'cr_boundary' layer to open the Symbology pane

- a. Choose the Black Outline (2pt) option in the Gallery
 - b. Then switch to the Properties tab and under Appearance, change the Outline Width to 3 pt
4. In the Contents Pane under 'Standalone Tables', right click CR_helotiales.csv and from 'Create Points From Table' click 'XY Table to Point'
 - a. Input table should be CR_helotiales.csv
 - b. For the Output Feature Class, click the folder icon to browse your files, locate the project database and save the feature class there as 'helotia_dist'
 - c. The X Field should be longitude and the Y Field should be latitude
 - d. Don't change the CS
5. In the Geoprocessing Pane, search for the 'Delete Field' tool from the Data Management toolbox
 - a. Set the Input Table as helotia_dist
 - b. Then under Methods, use the down arrow to choose 'Keep Fields'
 - c. Under fields check the boxes for id, url, latitude, longitude, scientific_name, and common_name
 - d. Click 'Run'
6. To symbolize the basins, click on the color box underneath the hydrobasins layer to open the Symbology for the layer
 - a. Click the circled back arrow above the 'Gallery' tab to go to the Primary symbology section.
 - b. Right underneath the section name use the drop down arrow to choose Graduated Color symbology instead of Single Symbol
 - c. Change the Field to SUB_AREA with 5 classes
 - d. Choose the color scheme Cyan-Light to Blue-Dark

7. To symbolize the mushroom point data, click on the dot symbol underneath the `auricularia_dist` layer to open the layer's Symbology pane
 - a. From the Gallery, choose Tear Pin 1 (40%)
 - b. Switch to the Properties tab
 - c. Under the Appearance ribbon, choose the Medium Fuschia color
 - d. For the outline color, choose the same medium shade of gray as ur other symbolization of auricularia distribution, and make the Outline Width 1pt
 - e. Click 'Apply'

Add anew field

8. Right click the `helotia_dist` layer and open the attribute table
9. Next to 'Field' at the top of the attribute table, click the 'Add' button to add the new field
 - a. Name the new field "`fungus_order`" and NOT "order" as it is apparently an illegal word in ArcGIS Pro
 - b. Then make sure that the Data Type is text
 - c. Go to the top of the screen and choose 'Save' within the Manage Edits ribbon
 - d. Exit out of the Fields tab
10. In the regular attribute table tab, find the new field you created
 - a. Right click the top/the field title where is says '`fungus_order`' and choose 'Calculate Field'
 - b. In the Calculate Field tool go to the text box underneath "`fungus_order =`" and write the following exactly
 - i. 'Helotiales' (include the apostrophes!!)
 - c. Click 'Apply' to make sure it puts Helotiales into each box underneath the new field, and then 'OK'

Use the Summarize Within tool

11. Open the Geoprocessing pane and search for the Summarize Within tool from the Analysis toolbox
 - a. For the Input Polygons, use the dropdown arrow to select the hydrobasins layer
 - b. Select the helotia_dist layer for the Input Summary Features
 - c. Save the Output Feature Class to the project geodatabase and name it 'helotia_by_basin'
 - d. Use the dropdown arrow to select fungal_order as the group field
 - e. Check the box to add group percentages
 - f. Click 'Run'
12. Once the helotia_by_basin layer has successfully loaded into the Contents pane, turn off the hydrobasins layer
 - a. Furthermore, drag the hydrobasins layer below the cr_boundary layer for nicer organization

Project everything mapped so far

13. In the Content Pane, right click the map (top of the pane if listed by drawing order) and go to Properties.
 - a. Find the Coordinate Systems tab, then in the search box, type "SIRGAS 2000 UTM Zone 16" then press 'Enter' on the keyboard
 - b. Make sure that you select the same projection typed, then click on the star next to it in order to save it to Favorites
 - c. Click 'OK'
14. Now search for and open the Project tool in the Geoprocessing Pane
 - a. For Input Dataset or Feature Class, use the dropdown arrow to select the cr_boundary layer

- b. Click on the folder icon next to Output Dataset or Feature Class, then name the file 'helotia_boundary' and save it in the project geodatabase
 - c. Use the dropdown arrow to select Current Map for the Output Coordinate System
 - d. Keep the transformation, then click 'Run'
 - e. Once the newly projected layer successfully loads into the Contents Pane, remove the original layer 'cr_boundary', but leave the Project tool open
- 15. In the Project tool, change the Input Dataset to helotia_by_basin and name the Output Dataset 'prj_hbb'
 - a. Make sure that the Output CS is to be saved in the project geodatabase, then click 'Run'
 - b. Once the newly projected layer successfully loads into the Contents Pane, remove the original layer 'helotia_by_basin' and 'hydrobasins'
- 16. In the Project tool, change the Input Dataset to helotia_dist and name the Output Dataset 'prj_helotia'
 - a. Make sure that the Output CS is to be saved in the project geodatabase, then click 'Run'
 - b. Once the newly projected layer successfully loads into the Contents Pane, remove the original layer 'helotia_dist'
- 17. Turn off the World Hillshade layer

Separate the hydrobasin polygons with the highest counts of Helotiales observed

- 18. In the Contents pane, right click the prj_hbb layer and open the attribute table
 - a. Select rows 3, 13, 15, 17, and 20
- 19. At the top of the screen, click Export Features from within the Export subribbon
 - a. For Input Features, prj_hbb should already be selected
 - b. 'Use the selected records: 5' should also be selected

- c. Save the Output Feature Class to the project geodatabase and name it 'dense_hbasins'
- d. Click 'OK'

Symbolize the new layer

- 20. Right click the new layer in the Contents pane and choose Symbology
 - a. Change the symbology to Single Symbol
 - b. Click the color box next to Symbol and go to the Gallery tab
 - c. Choose the Black Outline (1pt), then go back to the Properties tab
 - d. Change the Outline Color to the Poinsettia Red color and change the outline to 1.5pt
 - e. Click 'Apply'

Project the new layer

- 21. Now search for and open the Project tool in the Geoprocessing Pane
 - a. For Input Dataset or Feature Class, use the dropdown arrow to select the dense_hbasins layer
 - b. Click on the folder icon next to Output Dataset or Feature Class, then name the file 'prj_dense_hbasins' and save it in the project geodatabase
 - c. Use the dropdown arrow to select Current Map for the Output Coordinate System
 - d. Keep the transformation, then click 'Run'
 - e. Once the newly projected layer successfully loads into the Contents Pane, remove the original layer 'dense_hbasins'

- 22. Save!

Repeating all of the same steps again, but for the third fungal order, Xylariales

1. Go to Insert, then New Map to add a new map tab to the project
 - a. Title this map 'Xylariales'
2. In the Catalog pane, go to the project geodatabase under the Databases folder and add the following layers
 - a. cr_boundary
 - b. hydrobasins
2. Still in the the Catalog pane, add the mushroom distribution table, 'CR_xylaria.csv'
 - a. This file should be located through this pathway Folders → EmilyLeva_Datasets → CRxylaria.csv → CR_xylaria.csv
3. To correctly symbolize the country boundary, click on the color box underneath the 'cr_boundary' layer to open the Symbology pane
 - a. Choose the Black Outline (2pt) option in the Gallery
 - b. Then switch to the Properties tab and under Appearance, change the Outline Width to 3 pt
 - c. Click 'Apply'
4. In the Contents Pane under 'Standalone Tables', right click CR_xylaria.csv and from 'Create Points From Table' click 'XY Table to Point'
 - a. Input table should be CR_xylaria.csv
 - b. For the Output Feature Class, click the folder icon to browse your files, locate the project database and save the feature class there as 'xylaria_dist'
 - c. The X Field should be longitude and the Y Field should be latitude
 - d. Don't change the CS
5. In the Geoprocessing Pane, search for the 'Delete Field' tool from the Data Management toolbox

- a. Set the Input Table as xylaria_dist
 - b. Then under Methods, use the down arrow to choose 'Keep Fields'
 - c. Under fields check the boxes for id, url, latitude, longitude, scientific_name, and common_name
 - d. Click 'Run'
6. To symbolize the basins, click on the color box underneath the hydrobasins layer to open the Symbolology for the layer
 - a. Click the circled back arrow above the 'Gallery' tab to go to the Primary symbology section.
 - b. Right underneath the section name use the drop down arrow to choose Graduated Color symbology instead of Single Symbol
 - c. Change the Field to SUB_AREA with 5 classes
 - d. Choose the color scheme Cyan-Light to Blue-Dark
7. To symbolize the mushroom point data, click on the dot symbol underneath the xylaria_dist layer to open the layer's Symbolology pane
 - a. From the Gallery, choose Tear Pin 1 (40%)
 - b. Switch to the Properties tab
 - c. Under the Appearance ribbon, choose the Quetzal Green color
 - d. For the outline color, choose the same medium shade of gray (as ur other symbolizations of auricularia and helotia distributions) and make the Outline Width 1pt
 - e. Click 'Apply'

Add a new field

8. Right click the xylaria_dist layer and open the attribute table
9. Next to 'Field' at the top of the attribute table, click the 'Add' button to add the new field

- a. Name the new field “fungal_order” and NOT “order” as it is apparently an illegal word in ArcGIS Pro
 - b. Then make sure that the Data Type is text
 - c. Go to the top of the screen and choose ‘Save’ within the Manage Edits ribbon
 - d. Exit out of the Fields tab
10. In the regular attribute table tab, find the new field you created
- a. Right click the top/the field title where it says ‘fungal_order’ and choose ‘Calculate Field’
 - b. In the Calculate Field tool go to the text box underneath “fungal_order =” and write the following exactly
 - i. ‘Xylariales’ (include the apostrophes!!)
 - c. Click ‘Apply’ to make sure it puts Xylariales into each box underneath the new field, and then ‘OK’

Use the Summarize Within tool

11. Open the Geoprocessing pane and search for the Summarize Within tool from the Analysis toolbox
- a. For the Input Polygons, use the dropdown arrow to select the hydrobasins layer
 - b. Select the xylaria_dist layer for the Input Summary Features
 - c. Save the Output Feature Class to the project geodatabase and name it ‘xylaria_by_basin’
 - d. Use the dropdown arrow to select fungal_order as the group field
 - e. Check the box to add group percentages
 - f. Click ‘Run’
12. Once the xylaria_by_basin layer has successfully loaded into the Contents pane, turn off the hydrobasins layer

- a. Furthermore, drag the hydrobasins layer below the cr_boundary layer for nicer organization

Project everything mapped so far

13. In the Content Pane, right click the map (top of the pane if listed by drawing order) and go to Properties.

- a. Find the Coordinate Systems tab, then in the search box, type "SIRGAS 2000 UTM Zone 16" then press 'Enter' on the keyboard
- b. Make sure that you select the same projection typed, then click on the star next to it in order to save it to Favorites
- c. Click 'OK'

14. Now search for and open the Project tool in the Geoprocessing Pane

- a. For Input Dataset or Feature Class, use the dropdown arrow to select the cr_boundary layer
- b. Click on the folder icon next to Output Dataset or Feature Class, then name the file 'xylaria_boundary' and save it in the project geodatabase
- c. Use the dropdown arrow to select Current Map for the Output Coordinate System
- d. Keep the transformation, then click 'Run'
- e. Once the newly projected layer successfully loads into the Contents Pane, remove the original layer 'cr_boundary', but leave the Project tool open

15. In the Project tool, change the Input Dataset to xylaria_by_basin and name the Output Dataset 'prj_xbb'

- a. Make sure that the Output CS is to be saved in the project geodatabase, then click 'Run'
- b. Once the newly projected layer successfully loads into the Contents Pane, remove the original layer 'xylaria_by_basin' and 'hydrobasins'

16. In the Project tool, change the Input Dataset to xylaria_dist and name the Output Dataset 'prj_xylaria'

- a. Make sure that the Output CS is to be saved in the project geodatabase, then click 'Run'
- b. Once the newly projected layer successfully loads into the Contents Pane, remove the original layer 'xylaria_dist'

Clean up

17. Turn off the World Hillshade layer
18. Open the attribute table for the layer 'prj_xylaria' and select the following rows 11, 13, 25, 56, and 110
 - a. Go to the Geoprocessing pane and find the Delete Rows tool
 - b. For Input Rows select the 'prj_xylaria' layer
 - c. Toggle on the Use the Selected Records
 - i. Should say for 5 records if you selected the right rows/points
 - d. Click 'Run'

Separate the hydrobasin polygons with the highest counts of Xylariales observed

19. In the Contents pane, right click the prj_xbb layer and open the attribute table
 - a. Select rows 3, 13, 17, 20, 27
20. At the top of the screen, click Export Features from within the Export subribbon
 - a. For Input Features, prj_xbb should already be selected
 - b. 'Use the selected records: 5' should also be selected
 - c. Save the Output Feature Class to the project geodatabase and name it 'dense_xbasins'
 - d. Click 'OK'

Symbolize the new feature layer

21. Right click the new layer in the Contents pane and choose Symbology
 - a. Change the symbology to Single Symbol
 - b. Click the color box next to Symbol and go to the Gallery tab
 - c. Choose the Black Outline (1pt), then go back to the Properties tab
 - d. Change the Outline Color to the Poinsettia Red color and change the outline to 1.5pt
 - e. Click 'Apply'

Project the new layer

22. Now search for and open the Project tool in the Geoprocessing Pane
 - a. For Input Dataset or Feature Class, use the dropdown arrow to select the dense_xbasins layer
 - b. Click on the folder icon next to Output Dataset or Feature Class, then name the file 'prj_dense_xbasins' and save it in the project geodatabase
 - c. Use the dropdown arrow to select Current Map for the Output Coordinate System
 - d. Keep the transformation, then click 'Run'
 - e. Once the newly projected layer successfully loads into the Contents Pane, remove the original layer 'dense_xbasins'

23. Save!